

Part 3 — Detailed Solutions

1) Likelihood of the per-arm rewards (6 pts)

We want to compute the likelihood of the rewards observed for arm a up to time t :

$$p(R_{I_t(a)} \mid \theta_a, I_t(a)) = \prod_{f_i \in I_t(a)} \theta_a^{R_{f_i}} (1 - \theta_a)^{1 - R_{f_i}} = \theta_a^{S_t(a)} (1 - \theta_a)^{F_t(a)}.$$

Explanation: Since $R_{f_i} \sim \text{Bernoulli}(\theta_a)$ independently, the joint pmf factors as above. The exponent of θ_a is the number of successes $S_t(a)$, and the exponent of $(1 - \theta_a)$ is the number of failures $F_t(a) = N_t(a) - S_t(a)$.

2) Sufficiency and minimal sufficiency (4 pts)

By the Fisher–Neyman factorization theorem, from the likelihood we have

$$p(R_{I_t(a)} \mid \theta_a, I_t(a)) = \underbrace{\theta_a^{S_t(a)} (1 - \theta_a)^{N_t(a) - S_t(a)}}_{g(S_t(a), N_t(a); \theta_a)} \cdot \underbrace{1}_{h(R_{I_t(a)})}.$$

Thus, $(S_t(a), N_t(a))$ is sufficient for θ_a .

If $N_t(a)$ is fixed, the likelihood ratio of two samples depends on θ_a only through the difference in success counts. This is constant in θ_a iff the success counts agree. Hence $S_t(a)$ is minimal sufficient when $N_t(a)$ is fixed.

3) Beta–Bernoulli conjugacy (5 pts)

Assume the prior

$$\theta_a \sim \text{Beta}(\bar{\alpha}_a, \bar{\beta}_a), \quad p(\theta_a) \propto \theta_a^{\bar{\alpha}_a - 1} (1 - \theta_a)^{\bar{\beta}_a - 1}.$$

The likelihood is

$$p(R_{I_t(a)} \mid \theta_a) \propto \theta_a^{S_t(a)} (1 - \theta_a)^{F_t(a)}.$$

Multiplying prior and likelihood gives the posterior:

$$p(\theta_a \mid R_{I_t(a)}) \propto \theta_a^{(\bar{\alpha}_a - 1) + S_t(a)} (1 - \theta_a)^{(\bar{\beta}_a - 1) + F_t(a)}.$$

Therefore,

$$\theta_a \mid R_{I_t(a)}, I_t(a) \sim \text{Beta}(\alpha_t(a), \beta_t(a)),$$

where

$$\alpha_t(a) = \bar{\alpha}_a + S_t(a), \quad \beta_t(a) = \bar{\beta}_a + F_t(a).$$

4) Sequential update rule (3 pts)

Suppose at time t arm a is pulled and reward $R_t \in \{0, 1\}$ is observed. Then:

$$\alpha_{t+1}(a) = \alpha_t(a) + R_t, \quad \beta_{t+1}(a) = \beta_t(a) + (1 - R_t).$$

For any other arm $b \neq a$, the parameters remain unchanged:

$$\alpha_{t+1}(b) = \alpha_t(b), \quad \beta_{t+1}(b) = \beta_t(b).$$

Initialization: Before any pulls,

$$\alpha_1(a) = \bar{\alpha}_a, \quad \beta_1(a) = \bar{\beta}_a.$$

5) Posterior mean and MAP (2 pts)

For $\theta_a \sim \text{Beta}(\alpha_t(a), \beta_t(a))$:

Posterior mean.

$$\mathbb{E}[\theta_a \mid \text{data}] = \frac{\alpha_t(a)}{\alpha_t(a) + \beta_t(a)} = \frac{\bar{\alpha}_a + S_t(a)}{\bar{\alpha}_a + \bar{\beta}_a + N_t(a)}.$$

Posterior MAP. If $\alpha_t(a) > 1$ and $\beta_t(a) > 1$, the interior mode is

$$\text{mode}(\theta_a \mid \text{data}) = \frac{\alpha_t(a) - 1}{\alpha_t(a) + \beta_t(a) - 2}.$$

Edge cases:

- If $\alpha_t(a) \leq 1$, $\beta_t(a) > 1$: mode at 0.
- If $\alpha_t(a) > 1$, $\beta_t(a) \leq 1$: mode at 1.
- If $\alpha_t(a) \leq 1$ and $\beta_t(a) \leq 1$: modes at both 0 and 1.